Ronak Pradeep

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Research Summary

My research focuses on **knowledge agents**: systems that search, read, and reason over large collections of structured and unstructured data to answer hard questions. I study these agents from both the data side and the model side, building **data environments** and training agents with **reinforcement learning**. I also develop **evaluation methods and benchmarks** for **retrieval-augmented generation (RAG)**, including **nugget-based assessment** and **LLM judges**, and I co-organize the **TREC RAG Track**, a community-wide benchmark. The two threads form one loop: how we measure a response becomes the reward an agent learns from.

Education

- **University of Waterloo** Waterloo, ON
  PhD - Computer Science, Advisor: Jimmy Lin
  - Thesis: *Democratizing and Modernizing Information Access: From Open Rerankers to Scalable RAG Evaluation*
  - Recipient of the Apple PhD Fellowship
  - Recipient of the Snowflake AI Research Collaborative Grant
  - Coursework: Differential Privacy, Optimization for Data Science, High Stakes Information Retrieval, Affective Computing, Deep Learning for Biotechnology
  - Teaching Assistant: Algorithms, Data Structures, Introduction to Computer Science II

- **University of Waterloo** Waterloo, ON
  BMath - Double Major in Computer Science and Combinatorics and Optimization
  - Graduate Level Coursework: Deep Reinforcement Learning, Randomized Algorithms, Formal Languages and Parsing (100%), Computational Vision, Statistical Learning, Dependent Types and Software Verification (100%)
  - Part of the Term Dean's Honours List; Graduate Level Coursework Average 96.5%

Experience

- **Databricks AI Research** Remote
  Research Scientist, w/ Michael Bendersky *Jun 2026 - Present*
  - Synthetic data and reinforcement learning for post-training information-seeking agents

- **Yupp AI** Waterloo, ON
  Research Scientist *Jan 2025 - Apr 2026*
  - Agents, judges, models, labelers, evaluations, and leaderboards

- **Apple - Siri Information Intelligence** Seattle, WA
  Research Intern, w/ Yunyao Li and Ihab Ilyas *May 2023 - Aug 2023*
  - Knowledge graphs and large language models

- **Google AI** Remote
  Student Researcher, w/ Donald Metzler, Kai Hui, and Vinh Tran *Feb 2022 - Feb 2023*
  - Scaling differentiable search indexes to large corpora and models

- **University of Waterloo** Waterloo, ON
  Undergraduate Researcher *Apr 2017 - Aug 2020*
  - Paragraph retrieval and ranking with Jimmy Lin and Rodrigo Nogueira
  - Reading comprehension tasks with Pascal Poupart
  - Deep biologically plausible vision models with Jeff Orchard

- **Montreal Institute for Learning Algorithms (MILA)** Montreal, QC
  Visiting Researcher, w/ Chris Pal and Jie Fu *May 2019 - Dec 2019*
  - Open-domain question answering and graph representation learning
- **ContextLogic** San Francisco, CA
  AI Research Intern *Jan 2018 - Apr 2018*
  - Neural title generation for e-commerce products and attribute categorization
  - Curated the iMaterialist Challenge for the FGVC workshop at CVPR 2018
- **Royal Bank of Canada** Toronto, ON
  Research Developer *Aug 2016 - Dec 2016*
  - Document ranking and question answering using dual embedding space and seq2seq models
- **University of Waterloo** Waterloo, ON
  Undergraduate Teaching Assistant for Math 136 - Linear Algebra *Jan 2017 - Apr 2017*

Notable Professional Activities

- **TREC Retrieval-Augmented Generation (RAG) Track**
  Lead organizer (2024); co-organizer (2025–2026)
  - Built a realistic evaluation benchmark for scalable, diverse, and reproducible RAG systems
  - Organized the 2024 and 2025 tracks with Shivani Upadhyay, Nandan Thakur, Daniel Campos, Ian Soboroff, Nick Craswell, and Jimmy Lin
  - Supervising Nour Jedidi, Lingwei Gu, and Pouya Sadeghi for the 2026 track with Jimmy Lin

Publications

- **Auto-Judge: A Cross-Task Benchmark for Comparing LLM Judges for Citation-Grounded RAG** (SIGIR 2026)
  *Naghmeh Farzi, Tim Hagen, Eugene Yang, Maik Fröbe, **Ronak Pradeep**, Hossein A. Rahmani, Xi Wang, Oleg Zendel, Martin Potthast, Laura Dietz*
- **Automating Generation of Long-Form Queries** (SIGIR 2026)
  *Shivani Upadhyay, Daniel Campos, Nandan Thakur, **Ronak Pradeep**, Nick Craswell, Jimmy Lin*
- **Overview of the TREC 2025 Retrieval Augmented Generation (RAG) Track** (TREC 2025)
  *Shivani Upadhyay, Nandan Thakur, **Ronak Pradeep**, Nick Craswell, Daniel Campos, Jimmy Lin*
- **Assembling Your Personal AI Council in Yupp to Provide Multiple Perspectives** (WSDM 2026)
  *Jimmy Lin, **Ronak Pradeep**, Gilad Mishne, Pankaj Gupta*
- **NeuCLIRBench: A Modern Evaluation Collection for Monolingual, Cross-Language, and Multilingual Information Retrieval** (Under Review)
  *Dawn Lawrie, James Mayfield, Eugene Yang, Andrew Yates, Sean MacAvaney, **Ronak Pradeep**, Scott Miller, Paul McNamee, Luca Soldaini*
- **NeuCLIRTech: Chinese Monolingual and Cross-Language Information Retrieval Evaluation in a Challenging Domain** (Under Review)
  *Dawn Lawrie, James Mayfield, Eugene Yang, Andrew Yates, Sean MacAvaney, **Ronak Pradeep**, Scott Miller, Paul McNamee, Luca Soldaini*
- **Humanity's Last Exam** (Nature)
  *Long Phan, Alice Gatti, Ziwen Han, ..., **Ronak Pradeep**, ... [and other co-authors]*
- **RankLLM: A Python Package for Reranking with LLMs** (SIGIR 2025)
  *Sahel Sharifymoghaddam, **Ronak Pradeep**, Andre Slavescu, Ryan Nguyen, Andrew Xu, Zijian Chen, Yilin Zhang, Yidi Chen, Jasper Xian, Jimmy Lin*

- **Gosling Grows Up: Retrieval with Learned Dense and Sparse Representations Using Anserini** (SIGIR 2025)
  *Jimmy Lin, Arthur Haonan Chen, Carlos Lassance, Xueguang Ma, **Ronak Pradeep**, Tommaso Teofili, Jasper Xian, Jheng-Hong Yang, Brayden Zhong, Vincent Zhong*

- **Assessing Support for the TREC 2024 RAG Track: A Large-Scale Comparative Study of LLM and Human Evaluations** (SIGIR 2025)
  *Nandan Thakur, **Ronak Pradeep**, Shivani Upadhyay, Daniel Campos, Nick Craswell, Ian Soboroff, Hoa Trang Dang, Jimmy Lin*

- **Search Arena Meets Nuggets: Towards Explanations and Diagnostics in the Evaluation of LLM Responses** (ICTIR 2026)
  *Sahel Sharifymoghaddam, Shivani Upadhyay, Nandan Thakur, **Ronak Pradeep**, Jimmy Lin*

- **The Great Nugget Recall: Automating Fact Extraction and RAG Evaluation with Large Language Models** (SIGIR 2025)
  ***Ronak Pradeep**, Nandan Thakur, Shivani Upadhyay, Daniel Campos, Nick Craswell, Jimmy Lin*

- **Accelerating Listwise Reranking: Reproducing and Enhancing FIRST** (SIGIR 2025)
  *Zijian Chen, **Ronak Pradeep**, Jimmy Lin*

- **Initial Nugget Evaluation Results for the TREC 2024 RAG Track with the AutoNuggetizer Framework** (arXiv)
  ***Ronak Pradeep**, Nandan Thakur, Shivani Upadhyay, Daniel Campos, Nick Craswell, Jimmy Lin*

- **A Large-Scale Study of Relevance Assessments with Large Language Models Using UMBRELA** (ICTIR 2025)
  *Shivani Upadhyay, **Ronak Pradeep**, Nandan Thakur, Daniel Campos, Nick Craswell, Ian Soboroff, Hoa Trang Dang, Jimmy Lin*

- **Ragnarök: A Reusable RAG Framework and Baselines for TREC 2024 Retrieval-Augmented Generation Track** (ECIR 2025)
  ***Ronak Pradeep**, Nandan Thakur, Sahel Sharifymoghaddam, Eric Zhang, Ryan Nguyen, Daniel Campos, Nick Craswell, Jimmy Lin*

- **Prompts as Auto-Optimized Training Hyperparameters: Training Best-in-Class IR Models from Scratch with 10 Gold Labels** (arXiv)
  *Jasper Xian, Saron Samuel, Faraz Khoubsirat, **Ronak Pradeep**, Md Arafat Sultan, Radu Florian, Salim Roukos, Avirup Sil, Christopher Potts, Omar Khattab*

- **UMBRELA: UMbrela is the (Open-Source Reproduction of the) Bing RELevance Assessor** (arXiv)
  *Shivani Upadhyay, **Ronak Pradeep**, Nandan Thakur, Nick Craswell, Jimmy Lin*

- **ConvKGYarn: Spinning Configurable and Scalable Conversational Knowledge Graph QA Datasets with Large Language Models** (EMNLP 2024 Industry Track)
  ***Ronak Pradeep**, Daniel Lee, Ali Mousavi, Jeffrey Pound, Yisi Sang, Jimmy Lin, Ihab Ilyas, Saloni Potdar, Mostafa Arefiyan, Yunyao Li*

- **Entity Disambiguation via Fusion Entity Decoding** (NAACL 2024)
  *Junxiong Wang, Ali Mousavi, Omar Attia, **Ronak Pradeep**, Saloni Potdar, Alexander Rush, Umar Farooq Minhas, Yunyao Li*

- **Zero-Shot Cross-Lingual Reranking with Large Language Models for Low-Resource Languages** (ACL 2024)
  *Mofetoluwa Adeyemi, Akintunde Oladipo, **Ronak Pradeep**, Jimmy Lin*

- **LiT and Lean: Distilling Listwise Rerankers into Encoder-Decoder Models** (ECIR 2025)
  *Manveer Singh Tamber, **Ronak Pradeep**, Jimmy Lin*

- **RankZephyr: Effective and Robust Zero-Shot Listwise Reranking is a Breeze!** (arXiv)
  ***Ronak Pradeep**, Sahel Sharifymoghaddam, Jimmy Lin*
- **RankVicuna: Zero-Shot Listwise Document Reranking with Open-Source Large Language Models** (arXiv)
  Ronak Pradeep, Sahel Sharifymoghaddam*, Jimmy Lin*
- **Vector Search with OpenAI Embeddings: Lucene Is All You Need** (WSDM 2024 Demo)
  *Jimmy Lin, **Ronak Pradeep**, Tommaso Teofili, Jasper Xian*
- **How Does Generative Retrieval Scale to Millions of Passages?** (EMNLP 2023, SIGIR 2023 GenIR Workshop)
  Ronak Pradeep, Kai Hui*, Jai Gupta, Adam D Lelkes, Honglei Zhuang, Jimmy Lin, Donald Metzler, Vinh Q Tran**
- **naverloo @ TREC Deep Learning and NeuCLIR 2023: As Easy as Zero, One, Two, Three — Cascading Dual Encoders, Mono, Duo, and Listo for Ad-Hoc Retrieval** (TREC 2023)
  *Carlos Lassance, **Ronak Pradeep**, Jimmy Lin*
- **Towards Automated End-to-End Health Misinformation Free Search with a Large Language Model** (ECIR 2024)
  ***Ronak Pradeep**, Jimmy Lin*
- **Zero-Shot Listwise Document Reranking with a Large Language Model** (arXiv)
  *Xueguang Ma, Xinyu Zhang, **Ronak Pradeep**, Jimmy Lin*
- **ReadProbe: A Demo of Retrieval-Enhanced Large Language Models to Support Lateral Reading** (arXiv)
  *Dake Zhang, **Ronak Pradeep***
- **Pre-Processing Matters! Improved Wikipedia Corpora for Open-Domain Question Answering** (ECIR 2023 Reproducibility)
  *Manveer Singh Tamber, **Ronak Pradeep**, Jimmy Lin*
- **PyGaggle: A Gaggle of Resources for Open-Domain Question Answering** (ECIR 2023 Reproducibility)
  ***Ronak Pradeep**, Haonan Chen, Lingwei Gu, Manveer Singh Tamber, Jimmy Lin*
- **Neural Query Synthesis and Domain-Specific Ranking Templates for Multi-Stage Clinical Trial Matching** (SIGIR 2022)
  ***Ronak Pradeep**, Yilin Li, Yuetong Wang, Jimmy Lin*
- **Document Expansions and Learned Sparse Lexical Representations for MS MARCO V1 and V2** (SIGIR 2022)
  Xueguang Ma, **Ronak Pradeep***, Rodrigo Nogueira, Jimmy Lin*
- **Squeezing Water from a Stone: A Bag of Tricks for Further Improving Cross-encoder Effectiveness for Reranking** (ECIR 2022 Reproducibility)
  ***Ronak Pradeep**, Yuqi Liu, Xinyu Zhang, Yilin Li, Andrew Yates, Jimmy Lin*
- **Another Look at DPR: Reproduction of Training and Replication of Retrieval** (ECIR 2022 Reproducibility)
  *Xueguang Ma, Kai Sun, **Ronak Pradeep**, Minghan Li, Jimmy Lin*
- **New Nails for Old Hammers: Anserini and Pyserini at TREC 2021** (TREC 2021 Proceedings)
  *Jimmy Lin, Haonen Chen, Chengcheng Hu, Sheng-Chieh Lin, Yilin Li, Xueguang Ma, **Ronak Pradeep**, Jheng-Hong Yang, Chuan-Ju Wang, Andrew Yates, Xinyu Zhang*

- **Vera: Prediction Techniques for Reducing Harmful Misinformation in Consumer Health Search** (SIGIR 2021)
  ***Ronak Pradeep**, Xueguang Ma, Rodrigo Nogueira, Jimmy Lin*
- **Chatty Goose: A Python Framework for Conversational Search** (SIGIR 2021 Demo)
  *Edwin Zhang, Sheng-Chieh Lin, Jheng-Hong Yang, **Ronak Pradeep**, Rodrigo Nogueira, Jimmy Lin*
- **Pyserini: A Python Toolkit for Reproducible Information Retrieval Research with Sparse and Dense Representations** (SIGIR 2021)
  *Jimmy Lin, Xueguang Ma, Sheng-Chieh Lin, Jheng-Hong Yang, **Ronak Pradeep**, Rodrigo Nogueira*
- **H_2oloo at TAC 2020: Epidemic Question Answering** (TAC 2020 Proceedings)
  Justin Borromeo, **Ronak Pradeep***, Jimmy Lin*
- **Exploring Listwise Evidence Reasoning with T5 for Fact Verification** (ACL 2021)
  Kelvin Jiang, **Ronak Pradeep***, Jimmy Lin*
- **H_2oloo at TREC 2020: When all you got is a Hammer... Deep Learning, Health Misinformation, and Precision Medicine** (TREC 2020 Proceedings)
  ***Ronak Pradeep**, Xueguang Ma, Xinyu Zhang, Hang Cui, Ruizhou Xu, Rodrigo Nogueira, Jimmy Lin*
- **Scientific Claim Verification with VerT5erini** (LOUHI 2021: The 12th International Workshop on Health Text Mining and Information Analysis colocated with EACL 2021)
  ***Ronak Pradeep**, Xueguang Ma, Rodrigo Nogueira, Jimmy Lin*
- **A Replication Study of Dense Passage Retriever** (arXiv)
  *Xueguang Ma, **Ronak Pradeep**, Kai Sun, Jimmy Lin*
- **Covidex: Neural Ranking Models and Keyword Search Infrastructure for the COVID-19 Open Research Dataset** (Scholarly Document Processing @ EMNLP 2020)
  *Edwin Zhang, Nikhil Gupta, Raphael Tang, Xiao Han, **Ronak Pradeep**, Kuang Lu, Yue Zhang, Rodrigo Nogueira, Kyunghyun Cho, Hui Fang, Jimmy Lin*
- **The Expando-Mono-Duo Design Pattern for Text Ranking with Pretrained Sequence-to-Sequence Models** (arXiv)
  ***Ronak Pradeep**, Rodrigo Nogueira, Jimmy Lin*
- **Document Ranking with a Pretrained Sequence-to-Sequence Model** (EMNLP 2020 Findings)
  *Rodrigo Nogueira, Zhiying Jiang, **Ronak Pradeep**, Jimmy Lin*
- **Foveated Down-Sampling Techniques** (CVIS 2020)
  *Parsa Torabian, **Ronak Pradeep**, Jeff Orchard, Bryan Tripp*

Accomplishments

- **TREC NeuCLIR 2023**: Top submission in a cross-lingual information retrieval track
- **SCIVER: Verifying Scientific Claims with Evidence (Scholarly Document Processing @ NAACL 2021)**: Top submission based on the primary metric
- **Fact Extraction and VERification (FEVER) - 1st (As of Jan 2021)**: State-of-the-art result on a widely used fact verification benchmark
- **TREC Clinical Trials 2021/2022**: Top submission in a clinical trial matching task
- **TREC Health Misinformation 2020/2021/2022**: Top submission in the ad hoc retrieval task of a health misinformation benchmark
- **TREC Deep Learning 2020/2023**: Top submission in the document ranking task of a large-scale retrieval benchmark
- **TREC-COVID 2020**: Best automatic run in rounds 4 and 5, and best feedback run in round 3
- **MS MARCO Document Ranking - 1st (As of Sep 2020)**: State-of-the-art result on a widely used neural document ranking benchmark

- **MS MARCO Passage Ranking - 1st (As of May 2020)**: State-of-the-art result on a widely used neural passage ranking benchmark
- **DiMarco Undergraduate Scholarship in Computational Rhetoric**: Awarded annually to a single student for academic achievement and demonstrated interest in computational rhetoric
- **Terminal AI - Winner**: Developed a heuristic-based AI game bot that placed 1st among teams of top Waterloo students and ranked 2nd globally among 15k players at the time of submission
- **Citadel Datathon - NYC**: Placed 2nd among teams from top universities in North America
- **HackPrinceton - Top 10**: Implemented a tool for sentiment analysis of Twitter and Guardian news using the VADER lexicon and encoder-decoder LSTMs, and visualized the resulting trends
- **University of Waterloo President's Scholarship of Distinction and Research Award**: Awarded for strong academic performance and research experience

Mentorship

Students mentored through my PhD and professional work: Vikram Chandramohan, Haonan (Arthur) Chen, Steven Chen, Yidi Chen, Hang Cui, Lily Ge, Lingwei Gu, Qing Guo, Zibo (Daniel) Guo, Nikhil Gupta, Xiao (Johnson) Han, Nour Jedidi, Yuxuan Ji, Faraz Khoubsirat, Nathan Kuissi, Sisi Li, Yilin (Larry) Li, Estella Liu, Yuqi Liu, Xueguang Ma, Jie (Stefan) Min, Ryan Nguyen, Pouya Sadeghi, Nima Sadri, Andre Slavescu, Yuvan Sooryakumar, Kai Sun, Manveer Singh Tamber, Raghav Vasudeva, Yuetong Wang, Alvis Wong, Jasper Xian, Andrew Xu, Edward Xu, Ruizhou (Richard) Xu, Yicheng (Kevin) Xu, Jeongseop (Patrick) Yi, Edwin Zhang, Eric Zhang, Lijia (Lizzy) Zhang, Brayden Zhong, Vincent Zhong, Catherine Zhou, Joe Zou

Service

Senior Program Committee:
SIGIR 2026
Workshop Organizer:
Eval4RAG at ECIR 2024
Reviewer:
ACM Transactions on the Web; SIGIR 2024–2025; ICTIR 2025; Gen-IR at SIGIR 2024; ACL Rolling Review 2026